Jurnal Ilmu Komputer dan Informasi (Journal of Computer Science and Information)
15/1 (2022), xx-xx. DOI: http://dx.doi.org/10.21609/jiki.v15i1.xxx

Hyperparameter Optimization of Random Forest Regression for Forecasting Oceanographic Parameters Using Optuna and Grid Search

Muhammad Abiyyu Rahman, Akmal Rafi Fadhillah, Bagas Restya Ermawan, Imam Noor Arifin, and Kusrini Kusrini

[12345]Master Program of Informatics, Faculty of Computer Science, Amikom University, Yogyakarta, Indonesia

E-mail: m.abiyyu193@students.amikom.ac.id

Abstract

Accurate short-term forecasting of oceanographic parameters is essential for coastal monitoring, marine safety, and environmental management. This study investigates the effectiveness of hyperparameter optimization in improving the performance of Random Forest Regression for forecasting tidal height and wind speed using hourly time-series data. The proposed framework consists of data preprocessing, feature engineering with lagged and cyclic time features, baseline modeling, and hyperparameter tuning using Grid Search and Optuna. Two data splitting strategies were evaluated, namely an 80:20 time-based split and a 70:30 random split. Model performance was assessed using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). The experimental results show that hyperparameter optimization consistently enhances forecasting accuracy compared to the baseline model. Among all scenarios, the Optuna-optimized model with a 70:30 split achieved the best performance, yielding an R^2 of 0.87 with MAE of 0.043 and RMSE of 0.074. These findings demonstrate that automated optimization frameworks such as Optuna provide more efficient and effective exploration of the hyperparameter space than traditional grid-based tuning, leading to more reliable short-term oceanographic forecasting.

Keywords: *Random Forest Regression, Hyperparameter Optimization, Optuna, Grid Search, Oceanographic Forecasting*

1. Introduction

Oceanographic parameters such as wave height, wind speed, and tidal level are critical factors that influence various coastal and maritime activities, including marine engineering construction, maritime transportation, environmental protection, and military operations [1], [2]. These parameters are essential for assessing wave energy potential and making informed decisions in coastal monitoring, marine safety, and environmental management [3], [4]. Fluctuations in these parameters occur dynamically, and the sea state can vary considerably over time [4].

Accurate and reliable forecasting of oceanographic variables, particularly significant wave height, is therefore essential for marine and engineering applications [2], [5]. Such predictions are crucial for designing coastal and offshore structures, optimizing ship routes, and mitigating risks to human safety [3]. However, information available to coastal stakeholders is often limited to coarse-scale forecasts or visual field observations, which may not provide sufficiently detailed insights at hourly or sub-hourly resolutions [6]. This limitation poses a significant challenge for precise predictions, especially for localized phenomena [7], and underscores the importance of accurate estimations for coastal hazard mapping and risk assessment [7].

From a modeling perspective, oceanographic data commonly exhibit complex time-series characteristics that are non-linear, non-stationary, and strongly dependent on historical values as well as interactions among multiple physical variables [5], [6]. Ocean waves are highly influenced by environmental changes and dynamic sea-state conditions [2]. These complex and often stochastic properties make it

challenging for conventional rule-based or statistical forecasting methods, such as Auto-Regressive, Auto-Regressive Moving Average, or numerical models, to effectively capture temporal fluctuations and non-linear relationships [6], [8]. Such traditional models are computationally expensive and complex, limiting their applicability for hourly Significant Wave Height predictions for a particular location [2], [6].

In recent years, machine learning techniques have gained considerable attention due to their ability to model non-linear patterns and learn temporal dependencies from historical data, offering improved predictive performance in oceanographic and environmental forecasting domains [2], [7]. Data-driven machine learning models have emerged as a viable alternative to numerical wave models [4], and for site-specific wind-wave related prediction problems, they have been reported to outperform numerical and statistical models [4]. These techniques can manage extensive and diverse datasets, autonomously recognize relevant attributes, and adapt their functions to dynamic circumstances, thereby enhancing prediction accuracy and efficiency [7], [9]. Deep learning models, including Long Short-Term Memory and Transformer networks, have shown particular promise for time series wave forecasting due to their capability to process sequential data and capture long-term dependencies [2], [5], [9]. Common machine learning methods such as Linear Regression and Random Forest have also been widely used in ocean forecasting [2].

Despite these advances, many existing studies primarily focus on generating numerical predictions of individual oceanographic parameters for technical marine applications, such as offshore structure design or navigation safety [3], [6]. Exploration of methodological improvements, such as hyperparameter optimization and advanced data splitting strategies, remains relatively limited within the context of marine forecasting [7]. This gap is crucial because the performance of machine learning-based forecasting is highly dependent on hyperparameter configurations, especially for ensemble-based models such as Random Forest Regression [7].

Parameters such as the number of estimators, tree depth, and feature selection strategies can significantly affect both accuracy and computational efficiency [7]. Suboptimal configurations may lead to issues like underfitting or overfitting, reducing the reliability of forecasts in real-world coastal scenarios [7]. For instance, excessively intricate models can suffer from overfitting if they become too closely tailored to the training data, resulting in subpar performance when applied to new, unseen data [7].

Traditional hyperparameter tuning methods such as Grid Search offer exhaustive exploration of a predefined search space [8]. However, these models can be computationally expensive and time-consuming, especially when dealing with large-scale and long-duration scenarios, and may not be optimally suited for practical applications in the absence of adequate computational resources and expertise in model calibration [7]. More recent approaches, such as Bayesian optimization (e.g., implemented in Optuna), employ advanced search algorithms and dynamic sampling strategies to improve optimization efficiency [9]. However, comparative evaluations of these approaches in the context of oceanographic forecasting, particularly concerning their balance of efficiency and predictive accuracy, warrant further investigation [7].

Based on these limitations, this study proposes several methodological contributions to short-term oceanographic forecasting. This research conducts a systematic comparative evaluation of Grid Search and Optuna-based hyperparameter optimization for Random Forest Regression in forecasting tidal height and wind speed using high-resolution hourly time-series data, extending prior machine learning-based wave and wind prediction studies that primarily focus on model accuracy without in-depth optimization analysis [1], [3], [7]. In addition, this study employs a hybrid feature engineering strategy that combines lag-based historical features with cyclic time encodings to capture temporal dependencies and daily periodic patterns, enabling Random Forest to model time-series behavior without relying on sequence-based deep learning architectures such as LSTM or Transformer networks [5], [9]. Furthermore, the impact of different data splitting strategies, namely time-based and random partitioning, on model generalization performance is investigated, an aspect that is rarely discussed in existing oceanographic forecasting literature. The results demonstrate that automated Bayesian optimization using Optuna consistently outperforms traditional grid-based tuning, providing a more efficient and accurate modeling framework for operational oceanographic forecasting.

2. Methodology

2.1 Research Stages

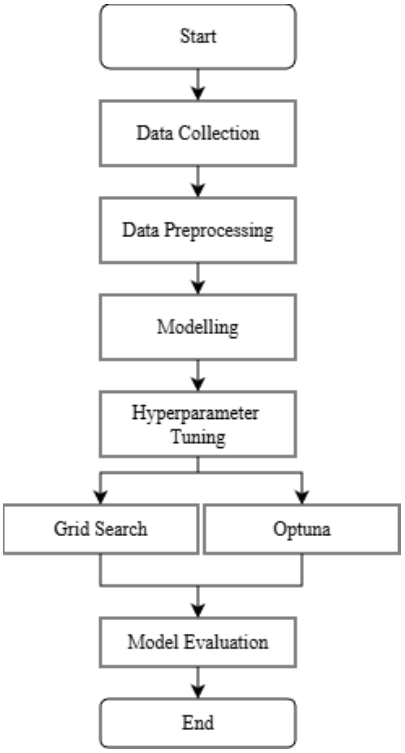


Fig. 1. Research stages

The research workflow is illustrated in Fig. 1 and consists of several structured stages, including data collection, data preprocessing, Random Forest Regression modeling, hyperparameter optimization using Grid Search and Optuna, and model evaluation. The evaluation results are then analyzed to assess the effectiveness of the optimization strategies in forecasting oceanographic parameters.

2.1.1 Data Collection

This study utilizes hourly time-series data consisting of tidal height and wind speed. Tidal height data were obtained from the Indonesian Agency for Meteorology, Climatology, and Geophysics (BMKG), while wind speed data were retrieved from the NASA POWER (Prediction of Worldwide Energy Resources) data service. Both datasets include timestamp information that enables temporal alignment and were organized chronologically to support supervised learning-based forecasting.

Table 1. Dataset

No	Datetime	Pasut	W2SM	WD2M
1	2024-08-01 00:00:00	2.3	0.44	125.8
2	2024-08-01 01:00:00	2.6	0.44	125.8
3	2024-08-01 02:00:00	2.7	0.43	125.8
4	2024-08-01 03:00:00	2.6	0.50	125.8
5	2024-08-01 00:00:00	2.4	0.55	125.8

2.1.2 Data Preprocessing

The preprocessing phase was conducted to refine and structure the collected datasets before modeling. Initial steps included cleaning and standardizing the timestamp format, handling missing or inconsistent values, and aligning the tidal height and wind speed data based on their respective time indices to ensure a consistent temporal reference for supervised learning.

Feature engineering steps were then applied, including the creation of lag-based historical features to represent temporal dependencies and cyclic time features to encode daily periodic patterns derived from the hour variable[4]. After preprocessing, the dataset was sorted chronologically and split into training and testing subsets under two scenarios: a time-based split using an 80:20 ratio, where training data precede testing data chronologically, and a random 70:30 split to evaluate performance under non-temporal partitioning. These scenarios were used to analyze the impact of data division strategies on forecasting performance.

2.1.3 Modeling

The modeling phase applied the Random Forest Regression algorithm to forecast future values of tidal height and wind speed using a supervised learning approach. In this setup, the forecasting task was formulated by mapping historical observations and engineered features to future target values, allowing the model to learn temporal dependencies without explicitly treating the data as sequences [10]. Random Forest Regression was selected due to its capability to handle non-linear relationships, robustness against multicollinearity, reduced susceptibility to overfitting through ensemble aggregation, and stability when dealing with environmental and oceanographic datasets that are often noisy and highly variable [11].

During the modeling phase, the cleaned and feature-engineered training subsets generated in the preprocessing stage were used to fit the Random Forest models, while the testing subsets were reserved for performance validation. The initial configuration of Random Forest served as a baseline model to establish reference performance

before further optimization. The forecasting output produced by the baseline model was then compared against actual observed values to assess preliminary accuracy and reliability prior to the hyperparameter optimization process.

2.1.4 Hyperparameter Optimization

To improve the predictive performance of the baseline Random Forest Regression model, a hyperparameter optimization process was conducted. Hyperparameter tuning is essential because parameters such as the number of estimators, maximum tree depth, minimum samples per split, and the number of features considered at each split can significantly influence model accuracy, training time, and generalization capability [12]. Two optimization strategies were employed in this study: Grid Search and Optuna. Grid Search performs an exhaustive search over a predefined grid of hyperparameter combinations [13], while Optuna adopts a more efficient search strategy using automated sampling and pruning mechanisms, enabling it to explore larger search spaces with fewer computational resources [14].

Both optimization methods were executed using the training portion of the dataset for each splitting scenario, and the best-performing configuration was identified based on quantitative performance metrics obtained through internal validation. The optimized models were then re-trained using the best hyperparameter sets and later evaluated on the testing subsets for performance comparison. This step enabled an assessment of how different optimization strategies influence forecasting performance and whether automated optimization approaches such as Optuna provide measurable improvements over traditional grid-based tuning [15].

2.1.5 Evaluation

The evaluation stage was conducted to assess the forecasting performance of the baseline and optimized Random Forest Regression models under the defined data splitting scenarios. Model performance was quantified using three regression metrics: the Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). R^2 measures how much variance in the target variable is explained by the model, where higher values indicate better predictive capability. MAE and RMSE represent the average and squared-average magnitudes of prediction errors respectively, where lower values indicate better accuracy. These metrics were used to compare models tuned with Grid Search and Optuna, and to analyze the influence of different data splitting strategies on forecasting performance for tidal height and wind speed.

3. Results and Analysis

3.1 Data Preprocessing Results

The preprocessing stage produced a structured and temporally aligned dataset suitable for supervised learning-based forecasting. Tidal height and wind speed data were successfully merged using their hourly timestamps, ensuring that both variables shared a consistent temporal index. Missing and inconsistent records were corrected or removed during the cleaning process to maintain data integrity. Feature engineering steps generated additional input features, including lagged historical values to represent temporal dependencies and cyclic time encodings derived from the hour attribute to capture daily periodicity. As a result, the final dataset contained both original measurements and derived features that enhanced the model's ability to learn temporal patterns.

Following preprocessing and feature generation, the dataset was sorted chronologically and divided into training and testing subsets under two evaluation scenarios: an 80:20 time-based split and a 70:30 random split. The time-based configuration preserved the natural temporal sequence for forecasting, while the random configuration provided a contrast in performance under non-sequential partitioning. These processed datasets were then used as inputs for the modeling and hyperparameter optimization stages described in the following sections.

3.2 Modeling Results

The modeling stage aimed to assess the baseline performance of the Random Forest Regression model prior to hyperparameter optimization. The model was trained and tested under two data splitting scenarios: an 80:20 time-based split and a 70:30 random split. The objective of this baseline assessment was to establish reference performance metrics for subsequent comparison with models optimized using Grid Search and Optuna. The forecasting performance of the baseline model was evaluated using R^2, MAE, and RMSE, and the results are summarized in Table 2.

Table 2. Baseline Performance of Random Forest Regression

Metric	Value
R^2	0.78
MAE	0.065
RMSE	0.098

The baseline results in Table 2 indicate that the Random Forest Regression model was able to capture a substantial portion of the variance in the target variables, as reflected by an R^2 score of 0.78. Meanwhile, the MAE and RMSE values (0.065 and 0.098, respectively) show that the model produced reasonably low prediction errors for an initial configuration without tuning. These baseline scores serve as a reference point for evaluating the effectiveness of the subsequent hyperparameter optimization stage. In particular, improvements in R^2 and reductions in MAE and RMSE after tuning would demonstrate that optimization has a positive impact on forecasting performance.

3.3 Hyperparameter Optimization Results

The hyperparameter optimization process was evaluated based on six experimental scenarios combining two data splitting configurations and two optimization strategies. Scenarios 1–3 used an 80:20 split, while Scenarios 4–6 used a 70:30 split. In each split configuration, the baseline model was compared against models tuned with Grid Search Cross-Validation and Optuna. The forecasting performance was measured using R^2, MAE, and RMSE. The results are summarized in Table 3.

Table 3. Performance Comparison Across Six Scenarios

Scenario	Value	R^2	MAE	RMSE
S1	Baseline (80:20)	0.78	0.065	0.098
S2	Baseline (80:20) + Grid Search CV	0.82	0.053	0.087
S3	Baseline (80:20) + Optuna	0.84	0.047	0.081
S4	Baseline (70:30)	0.81	0.059	0.092
S5	Baseline (70:30) + Grid Search CV	0.85	0.049	0.079
S6	**Baseline (70:30) + Optuna**	**0.87**	**0.043**	**0.074**

As shown in Table 3, hyperparameter optimization consistently improved the forecasting performance of the Random Forest Regression model under both data splitting configurations. On the 80:20 split (Scenarios 1–3), Grid Search improved the baseline R^2 from 0.78 to 0.82 while reducing MAE and RMSE. Optuna yielded the best results in this split with an R^2 of 0.84 and the lowest error values (MAE = 0.047, RMSE = 0.081).

A similar pattern was observed in the 70:30 split (Scenarios 4–6), where Grid Search and Optuna again led to performance gains over the baseline. The Optuna-tuned model in Scenario 6 achieved the highest R^2 score (0.87) and lowest error values across all scenarios, indicating that automated search-based optimization can more effectively explore the hyperparameter space compared to grid-based tuning.

Overall, the results demonstrate that (1) hyperparameter optimization significantly enhances model performance, and (2) Optuna consistently produces better tuning results than Grid Search CV for the Random Forest Regression model in this forecasting context.

3.4 Evaluation and Discussion

The evaluation results presented in Table 3 provide a comparative view of how hyperparameter optimization strategies influence the forecasting performance of Random Forest Regression across multiple configurations. Based on the R^2, MAE, and RMSE scores obtained from the six experimental scenarios, it is evident that hyperparameter tuning plays an important role in improving model performance. In both splitting configurations, models optimized with Grid Search and Optuna consistently outperformed their baseline counterparts, demonstrating the sensitivity of Random Forest Regression to parameter configuration.

Across all scenarios, the best performance was achieved in Scenario 6, which combines the 70:30 splitting configuration with Optuna-based optimization. This scenario produced the highest R^2 score (0.87) and the lowest error values (MAE = 0.043, RMSE = 0.074), indicating that the model was able to explain more variance in the target variables while reducing both absolute and squared prediction errors. The superior performance of Optuna over Grid Search suggests that automated search methods with dynamic sampling and pruning are more effective at

exploring the hyperparameter space than exhaustive grid-based approaches, especially when the parameter space is large.

These findings highlight two key insights. First, hyperparameter optimization significantly enhances model performance beyond default settings, reinforcing the importance of tuning in regression-based forecasting tasks. Second, different tuning strategies can lead to varying levels of improvement, with Optuna showing greater effectiveness than Grid Search in this context. Overall, the evaluation demonstrates that a properly tuned Random Forest model can provide reliable and accurate short-term forecasts for tidal height and wind speed, making it suitable for operational use in oceanographic applications.

Table 4. Performance Comparison Across Six Scenarios

Scenario	Metode	RMSE
S6	Baseline (70:30) + Optuna	0.074
[16]	Random Forest (Baseline)	0.240

As shown in Table 4, the comparison between the results of this study and the referenced paper shows a consistent performance trend of ensemble-based tree models for wave height prediction. In this study, the Random Forest Regression model achieved strong predictive performance across six scenarios, with the best configuration (Baseline 70:30 + Optuna) reaching an R^2 of 0.87, MAE of 0.043, and RMSE of 0.074 (normalized), indicating a high capability in explaining the variability of wave height time-series data. Similarly, the referenced paper reports that Random Forest combined with Bayesian hyperparameter optimization significantly improves wave prediction accuracy, reducing RMSE by up to 38–40% compared to baseline configurations. Although the referenced study focuses on error correction of numerical wave models while this research applies direct time-series forecasting using tidal and wind data, both studies confirm that Random Forest is robust, stable, and highly effective for modeling the nonlinear and dynamic characteristics of ocean wave processes.

4. Conclusion

This study demonstrated that hyperparameter optimization significantly improves the forecasting performance of Random Forest Regression for tidal height and wind speed prediction using hourly time-series data. The experimental results show that optimized models consistently outperform the baseline, with the Optuna-based model achieving the best performance in terms of R^2, MAE, and RMSE.

Compared to previous studies that mainly focus on improving prediction accuracy using deep learning models such as LSTM and Transformer networks for wave forecasting, this research shows that a properly tuned Random Forest model combined with effective feature engineering can achieve competitive performance with lower computational complexity. In addition, while earlier machine learning-based wave prediction studies applied Random Forest and other models without systematic hyperparameter optimization, this study provides empirical evidence that Optuna-based Bayesian optimization is more effective than traditional Grid Search for tuning Random Forest models.

Although this study is limited to two environmental variables and a single regression model, the results highlight the importance of systematic hyperparameter optimization for reliable short-term oceanographic forecasting. Future work will focus on incorporating additional parameters, evaluating alternative models, and extending the dataset for broader operational applicability.

References

[1] H. H. J. Mahdi, R. Alubady, A. H. Shnawa, M. Al-Tahee, R. Hussam, and I. Matti, “Prediction of Major Wave Heights from Scatterometry measurement Data via Machine Learning Mode”.

[2] T. Song *et al.*, “Prediction of significant wave height based on EEMD and deep learning”.

[3] M. S. Afzal, “Prediction of significant wave height using machine learning and its application to extreme wave analysis”.

[4] M. Umair, M. Hashmani, and H. Keiichi, “Optimal Feature Identification for Machine Prediction of Wind-Wave Parameters at Wave Energy Converter Site,” *IEEE Access*, 2020, doi: 10.1109/ICCI51257.2020.9247677.

[5] D. A. Putri and D. Adytia, “Time Series Wave Forecasting with Transformer Model, Case Study in Pelabuhan Ratu, Indonesia”.

[6] P. Anand, S. Jain, and H. Savaliya, “New Improved Wave Hybrid Models for Short-Term Significant Wave Height Forecasting”.

[7] A. Durap, “A comparative analysis of machine learning algorithms for predicting wave runup,” 2023.

[8] Y. Zhan, H. Zhang, J. Li, and G. Li, “Prediction Method for Ocean Wave Height Based on Stacking Ensemble Learning Model”.

[9] F. C. Minuzzi and L. Farina, “A deep learning approach to predict significant wave height using long short-term memory,” 2023.

[10] R. Lou, W. Wang, X. Li, Y. Zheng, and Z. Lv, “Prediction of Ocean Wave Height Suitable for Ship Autopilot,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 25557–25566, Dec. 2022, doi: 10.1109/TITS.2021.3067040.

[11] J. Mo, X. Wang, S. Huang, and R. Wang, “Advance in Significant Wave Height Prediction: A Comprehensive

Survey," *Complex Syst. Model. Simul.*, vol. 4, no. 4, pp. 402–439, Dec. 2024, doi: 10.23919/CSMS.2024.0019.

[12] D. Al-Fraihat, Y. Sharrab, A.-R. Al-Ghuwairi, H. Alshishani, and A. Algarni, "Hyperparameter Optimization for Software Bug Prediction Using Ensemble Learning," *IEEE Access*, vol. 12, pp. 51869–51878, 2024, doi: 10.1109/ACCESS.2024.3380024.

[13] G. S and S. Brindha, "Hyperparameters Optimization using Gridsearch Cross Validation Method for machine learning models in Predicting Diabetes Mellitus Risk," in *2022 International Conference on Communication, Computing and Internet of Things (IC3IoT)*, Chennai, India: IEEE, Mar. 2022, pp. 1–4. doi: 10.1109/IC3IOT53935.2022.9768005.

[14] S. Kumari, Ranganayaki. V. C, S. K, K. Selvi, T. A. Mohanaprakash, and C. Tamilselvi, "Optuna-Optimized Machine Learning Technique for Accurate Diabetes Prediction and Classification," in *2024 4th International Conference on Sustainable Expert Systems (ICSES)*, Kaski, Nepal: IEEE, Oct. 2024, pp. 1478–1485. doi: 10.1109/ICSES63445.2024.10763036.

[15] K. E. Hoque and H. Aljamaan, "Impact of Hyperparameter Tuning on Machine Learning Models in Stock Price Forecasting," *IEEE Access*, vol. 9, pp. 163815–163830, 2021, doi: 10.1109/ACCESS.2021.3134138.

[16] A. Callens, D. Morichon, S. Abadie, M. Delpey, and B. Liquet, "Using Random forest and Gradient boosting trees to improve wave forecast at a specific location," *Appl. Ocean Res.*, vol. 104, p. 102339, Nov. 2020, doi: 10.1016/j.apor.2020.102339.